9th Class • Punjab Board • PECTAA New Syllabus

Chapter 1: Introduction to Computational Systems

Complete solved exercise — MCQs, short questions, long answers, worked binary arithmetic, important questions and revision.

Chapter 1 • Complete official exercise • 10 MCQs + 11 short + 6 long questions

What this chapter is about

• A **system** is an organised collection of interdependent parts working towards a common goal. Its objective, components, environment and communication explain how it works.

• The **von Neumann architecture** stores both instructions and data in memory. The CPU repeatedly fetches, decodes, executes and stores results.

• Computers represent numbers and text in binary. This chapter covers number bases, binary arithmetic, signed integers, ASCII, Unicode and common storage units.

• **System software** manages the computer; **application software** helps users perform tasks such as writing, calculating and designing.

Nine-unit / twelve-unit coverage map

• Chapter 1 combines the subject matter printed separately as **Unit 1 Introduction to Systems**, **Unit 2 Number Systems** and **Unit 5 Software System** in the twelve-unit textbook.

• The exercise below follows the **April 2026 nine-unit PECTAA edition in its original order**. The twelve-unit text was used to cross-check concepts and wording, not to omit the merged software or number-system material.

A. Multiple Choice Questions

Select the correct option. Every answer includes a short reason.

1. The primary function of a system is:

(a) To work independently
(b) To achieve a common goal
(c) To create new systems
(d) To provide entertainment

Answer: (b) To achieve a common goal — A system is an organised set of parts that work together for one shared purpose or goal.

2. One of the fundamental concepts of any system is:

(a) Its size
(b) Its objective
(c) Its age
(d) Its price

Answer: (b) Its objective — The objective states what the system is meant to achieve; size, age and price do not define every system.

3. An example of a simple system is:

(a) A human body
(b) A computer network
(c) A thermostat regulating temperature
(d) The Internet

Answer: (c) A thermostat regulating temperature — A thermostat has a small, clear set of parts: a sensor, a controller and a heating or cooling output.

4. The basic components of a system are:

(a) Users, hardware, software
(b) Objectives, components, environment, communication
(c) Inputs, outputs, processes
(d) Sensors, actuators, controllers

Answer: (b) Objectives, components, environment, communication — The chapter describes a system through its objective, components, environment and communication among components.

5. Which of the following describes the Von Neumann architecture's main characteristic?

(a) Separate memory for data and instructions
(b) Parallel execution of instructions
(c) Single memory store for both program instructions and data
(d) Multiple CPUs for different tasks

Answer: (c) Single memory store for both program instructions and data — Von Neumann's stored-program idea keeps instructions and data in the same main memory.

6. ASCII stand for:

(a) American Standard Code for Information Interchange
(b) Advanced Standard Code for Information Interchange
(c) American Standard Communication for Information Interchange
(d) Advanced Standard Communication for Information Interchange

Answer: (a) American Standard Code for Information Interchange — ASCII is the American Standard Code for Information Interchange.

7. How many bits are used in the standard ASCII encoding?

(a) 7 bits
(b) 8 bit
(c) 16 bits
(d) 32 bits

Answer: (a) 7 bits — Standard ASCII uses 7 bits and therefore represents $2^7 = 128$ characters. Extended ASCII uses 8 bits.

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8. How many bytes are used to store a typical integer?

(a) 1 byte (b) 2 bytes
(c) 4 bytes (d) 8 bytes

Answer: (c) 4 bytes — The textbook uses 4 bytes (32 bits) for a typical integer. Actual programming languages also provide other integer sizes.

9. Which software is used to enhance system performance and security?

(a) Operating system (b) Utility software
(c) Application software (d) Device drivers

Answer: (b) Utility software — Utilities perform maintenance and protection tasks such as disk cleanup, backup and antivirus scanning.

10. Which of the following is an example of application software?

(a) Microsoft Word (b) BIOS
(c) Disk Cleanup (d) Device Manager

Answer: (a) Microsoft Word — Microsoft Word helps a user perform a specific task—word processing—so it is application software.

B. Short Questions

1. Define a system.

A system is an organised collection of interdependent parts that work together to perform a particular function or achieve a common goal. For example, a computer system combines hardware, software, data and users to process information.

2. Enlist the main components of the Von Neumann architecture?

Its main components are the **CPU** (control unit, arithmetic logic unit and registers), **main memory**, **input devices**, **output devices**, **secondary storage** and the **system bus** that carries data, addresses and control signals.

3. Enlist the four main steps in the Von Neumann architecture's instruction cycle?

The four steps are **fetch** the instruction from memory, **decode** it in the control unit, **execute** it using the ALU or another component, and **store** the result in a register or memory. The cycle then repeats for the next instruction.

4. What is a key advantage of the Von Neumann architecture?

Its stored-program design is **flexible**: a program can be changed by loading different instructions into memory without redesigning the hardware. Using one memory for instructions and data also simplifies the computer's design.

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5. What is the primary purpose of the ASCII encoding scheme?

ASCII gives each English letter, digit, punctuation mark and control character a standard numeric code so that computers can store and exchange text consistently. Standard ASCII uses 7 bits for 128 characters.

6. Define Unicode.

Unicode is an international character standard that assigns a unique code point to characters used in the world's writing systems, as well as symbols and emoji. UTF-8, UTF-16 and UTF-32 are different ways of encoding those code points in bytes.

7. How does the number of bits affect the range of integer values?

Each added bit doubles the number of possible bit patterns. An n-bit unsigned integer has 2^n patterns and stores values from 0 to $2^n - 1$. In two's-complement signed form, the range is -2^{n-1} to $2^{n-1} - 1$.

8. Define system software and provide two examples.

System software manages hardware and provides a platform on which applications run. Two examples are an **operating system**, such as Microsoft Windows or Linux, and **utility software**, such as Disk Cleanup or an antivirus program.

9. Differentiate between system software and application software.

System software	Application software
Manages the computer, hardware resources and basic services.	Helps the user perform a particular task or activity.
Provides a platform and user interface for other programs.	Runs on the platform supplied by the system software.
Examples: operating system, device driver, utility and firmware.	Examples: word processor, spreadsheet, browser and graphic-design program.
Usually works in the background and is essential for operating the machine.	Normally selected and started by the user for work, learning or entertainment.

10. Add 1100_2 and 1011_2.

Align the bits from the right and use $1 + 1 = 10_2$ (write 0, carry 1):

1 1 1 0 ← carries

1 1 0 0_2 = 12_{10}

+ 1 0 1 1_2 = 11_{10}

$$1\ 0\ 1\ 1\ 1_2 = 23_{10}$$

Answer: $1100_2 + 1011_2 = 10111_2$

11. Subtract 0011_2 from 1010_2.

Subtract each column, borrowing 2_2 from the next higher place when required:

$$1\ 0\ 1\ 0_2 = 10_{10}$$

$$-\ 0\ 0\ 1\ 1_2 = 3_{10}$$

$$0\ 1\ 1\ 1_2 = 7_{10}$$

Answer: $1010_2 - 0011_2 = 0111_2 = 111_2$

C. Long Questions

1. Describe the basic concept of a system.

A system is more than a random collection of objects. Its parts are **related and interdependent**, so each part contributes to the working of the whole. A change in one part can affect the other parts and the final result.

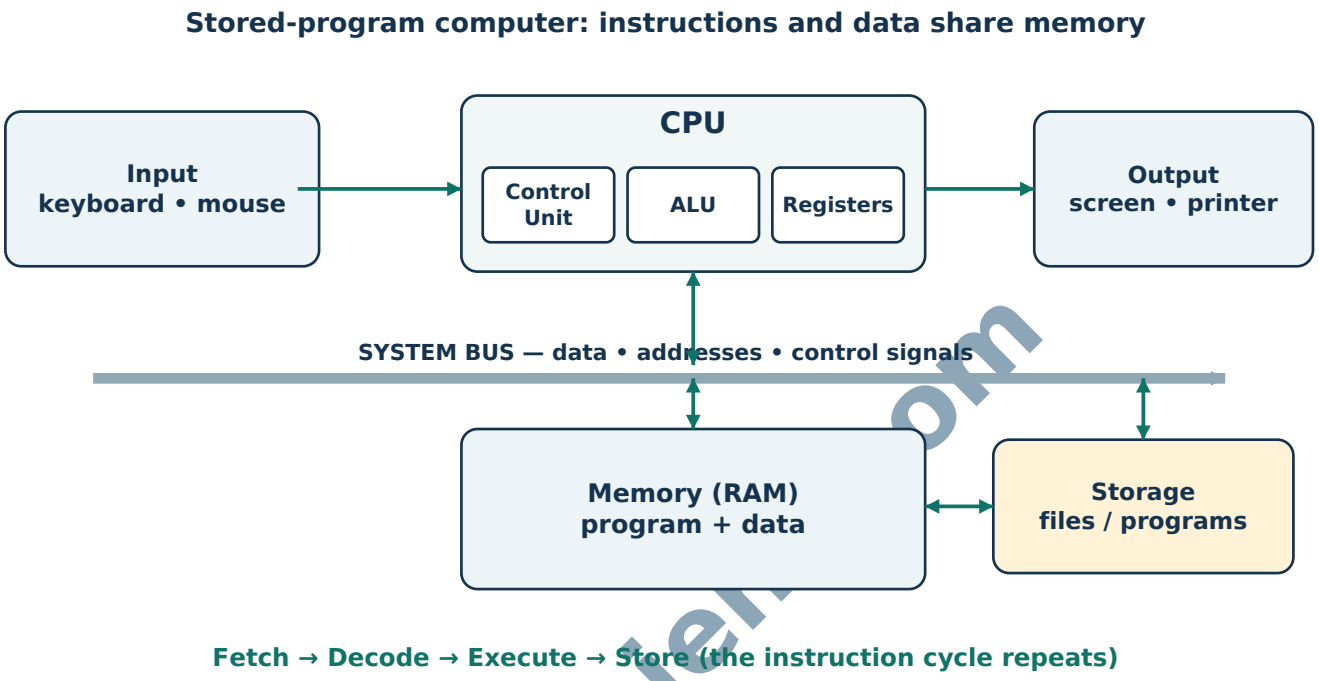
Feature	Meaning	Example: school computer system
Objective	The purpose or goal to be achieved.	Help students learn and perform digital tasks.
Components	The parts that perform different roles.	Computers, users, software, network and electricity.
Communication	The exchange of data or signals among components.	A student sends a document to a shared printer.
Environment	Outside conditions that affect or interact with the system.	Classroom, power supply, Internet service and school rules.
Interaction	The way components depend on and influence one another.	The operating system controls hardware so browser software can open a page.

Systems may be **natural**, such as the solar system or human body, or **artificial**, such as a bicycle, computer network or school. Understanding the objective and interactions helps us analyse faults and improve the system instead of looking at each part in isolation.

2. Explain the Von Neumann Architecture of a computer.

Von Neumann architecture is based on the **stored-program concept**. Program instructions and the data used by those instructions are kept in the same main memory. The CPU reads instructions one after another and performs the required operations.

Figure 1 — Main components and data paths



Component	Function
Control unit (CU)	Fetches and decodes instructions and coordinates the other components through control signals.
Arithmetic logic unit (ALU)	Performs arithmetic operations such as addition and logical operations such as comparison.
Registers	Very fast storage locations inside the CPU that temporarily hold instructions, addresses, operands and results.
Main memory (RAM)	Holds the current program instructions and data. Both share this memory in the basic model.
Input / output	Input brings data and commands into the computer; output presents processed results to the user or another system.
Secondary storage	Keeps programs and data permanently, even when power is switched off.
System bus	Transfers data, memory addresses and control signals among components.

Advantages: The design is relatively simple and flexible. A different program can be loaded without changing the hardware. **Limitation:** Instructions and data share a

communication path, so the CPU may have to wait; this is called the **von Neumann bottleneck**. Shared memory also requires protection against accidental or malicious alteration of instructions.

3. Explain the working of Von Neumann Architecture.

The computer works by repeating the **instruction cycle**. A program counter keeps the address of the next instruction, while registers hold the instruction and values currently being used.

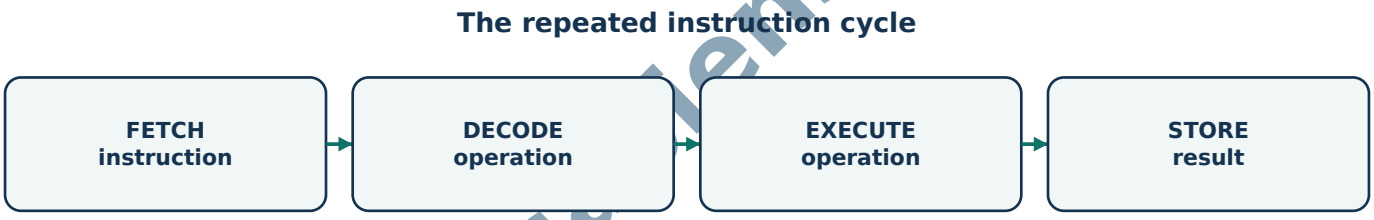
1. Fetch: The control unit places the next instruction's address on the address bus and reads that instruction from main memory into the CPU. The program counter then moves to the following instruction.

2. Decode: The control unit identifies the operation and determines which data, registers or memory addresses are needed.

3. Execute: The ALU or another component performs the operation—for example, adding two values, comparing them or moving data.

4. Store: The result is written to a register or main memory. Output may also be sent to a device. The next fetch then begins.

Figure 2 — Fetch-decode-execute-store



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Example: For an instruction meaning “add the value at memory location 20 to register R1”, the CPU fetches and decodes the instruction, reads the value from location 20, adds it to R1 in the ALU and stores the new sum in R1. Input, memory, storage and output communicate through the system bus while the control unit coordinates the steps.

4. Explain how characters are encoded using Unicode also provide examples.

Unicode assigns every character a unique **code point**, written in the form **U+ hexadecimal value**. A code point is an abstract number; an encoding form such as UTF-8 or UTF-32 converts it into bytes for storage and transmission.

Encoding	Storage pattern	Main feature
UTF-8	Uses 1–4 bytes per character.	Variable length; ASCII characters keep their original one-byte codes, so it is widely used on the web.

Encoding	Storage pattern	Main feature
UTF-16	Uses one or two 16-bit code units (2 or 4 bytes).	Common characters use 2 bytes; supplementary characters use a surrogate pair.
UTF-32	Uses a 32-bit code unit (4 bytes) for every character.	Fixed length and simple indexing, but usually needs more storage.

Unicode examples

Character	Unicode code point	Example encoding
A	U+0041	UTF-8 byte 41_{16}; UTF-16 code unit 0041_{16}; UTF-32 code unit 00000041_{16}.
ا (Arabic/Urdu alef)	U+0627	UTF-8 bytes $D8\ A7_{16}$, so this non-ASCII character needs two bytes.
😀	U+1F600	UTF-8 uses four bytes; UTF-32 uses the single code unit $0001F600_{16}$.

Thus, Unicode allows English, Urdu, Arabic, Chinese and many other scripts to appear in the same document. The receiving software must use the same encoding form; otherwise, the bytes may be decoded as incorrect characters.

5. Describe in detail how integers are stored in computer memory.

Integers are stored as fixed groups of bits. A group of n bits can form 2^n different patterns. The interpretation of those patterns depends on whether the value is unsigned or signed and on the chosen representation.

Unsigned n-bit range: 0 to $2^n - 1$

Two's-complement signed n-bit range: -2^{n-1} to $+2^{n-1} - 1$

8-bit unsigned range: 0 to 255 • 8-bit signed range: −128 to +127

Representation	How +13 and −13 appear in 8 bits	Key point
Unsigned	+13 = 00001101; negative values are not represented.	All n bits contribute to the magnitude.
Sign and magnitude	+13 = 00001101; −13 = 10001101.	Leftmost bit is the sign; remaining bits store the magnitude. It has two zeros.
One's complement	+13 = 00001101; −13 = 11110010.	Invert every bit of the positive value. It also has two zeros.

Representation	How +13 and −13 appear in 8 bits	Key point
Two's complement	+13 = 00001101; −13 = 11110011.	Invert every bit and add 1. It has one zero and makes binary addition/subtraction simpler.

Forming −13 in 8-bit two's complement

+13 = 00001101

Invert all bits = 11110010

Add 1 = 11110011

Check: 00001101 + 11110011 = 1 00000000; discard the 9th carry → 00000000

The leftmost bit of a signed value indicates its sign: 0 for non-negative and 1 for negative in these representations. A computer must also use the agreed width; for example, the textbook treats a typical integer as 32 bits (4 bytes).

6. Perform the following binary arithmetic operations:
(a) Multiplication of 101_2 by 11_2 (b) Division of 1100_2 by 10_2

(a) $101_2 \times 11_2$

Multiply by each bit of 11_2. The second partial product is shifted one place to the left, just as in decimal multiplication.

$1\ 0\ 1_2 = 5_{10}$

$\times\ 1\ 1_2 = 3_{10}$

$1\ 0\ 1 \leftarrow 101 \times 1$

$+\ 1\ 0\ 1\ 0 \leftarrow 101 \times 1$, shifted one place

$1\ 1\ 1\ 1_2 = 15_{10}$

Answer: $101_2 \times 11_2 = 1111_2$

(b) $1100_2 \div 10_2$

Binary long division gives quotient bits 1, 1 and 0. There is no remainder.

11_2 goes into 11_2 once → first quotient bit 1, remainder 01_2

Bring down 0: $10_2 \div 10_2 = 1$, remainder 0

Bring down the final 0: $0 \div 10_2 = 0$

Quotient = 110_2; remainder = 0_2

Decimal check: $1100_2 = 12$, $10_2 = 2$ and $12 \div 2 = 6 = 110_2$

Answer: $1100_2 \div 10_2 = 110_2$

D. More Important Questions (from the text)

1. Compare decimal, binary, octal and hexadecimal number systems. Convert 156_{10} into all three other bases.

System	Base	Allowed digits	Place values
Decimal	10	0–9	…, 10^2, 10^1, 10^0
Binary	2	0, 1	…, 2^2, 2^1, 2^0
Octal	8	0–7	…, 8^2, 8^1, 8^0
Hexadecimal	16	0–9, A–F	…, 16^2, 16^1, 16^0

Repeated division by 2 gives remainders (read upwards): 0,0,1,1,1,0,0,1 → 10011100_2

Repeated division by 8: $156 \div 8 = 19$ r4; $19 \div 8 = 2$ r3; $2 \div 8 = 0$ r2 → 234_8

Repeated division by 16: $156 \div 16 = 9$ r12(C); $9 \div 16 = 0$ r9 → $9C_{16}$

Checks: $234_8 = 2\times64 + 3\times8 + 4 = 156$; $9C_{16} = 9\times16 + 12 = 156$

$156_{10} = 10011100_2 = 234_8 = 9C_{16}$

2. How do application software packages help in business and education?

In business: Word processors prepare letters and reports; spreadsheets record sales, calculate costs and analyse budgets; presentation software supports meetings; communication tools connect staff and customers; database and accounting packages organise records. These tools reduce repeated manual work, improve accuracy and make information easier to share.

In education: Word processors and presentation tools support assignments and lessons; spreadsheets help analyse results; browsers and learning platforms give

access to educational material; graphic tools support creative projects; school-management software helps with attendance, timetables and communication with parents.

Application software therefore improves productivity, but it still depends on the operating system, suitable hardware, accurate data and responsible users.

3. How are images, audio and video stored, and what storage units are used?

Images: A digital image consists of pixels. Each pixel's colour is stored as numbers; in RGB, red, green and blue components commonly use values from 0 to 255. JPEG, PNG and GIF are common formats.

Audio: A sound wave is measured at regular intervals (**sampling**), and each sample is converted to a numeric level (**quantisation**). MP3, WAV and AAC are common formats. Video stores a rapid sequence of image frames together with audio; MP4, AVI and MKV are common formats.

Unit	Relationship
Bit (b)	Smallest binary unit: 0 or 1.
Byte (B)	8 bits.
Kilobyte (KB)	1024 bytes in the textbook's binary storage convention.
Megabyte (MB)	1024 KB
Gigabyte (GB)	1024 MB
Terabyte (TB)	1024 GB
Petabyte (PB)	1024 TB

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Points to Remember

System: interdependent components working together for a common objective; study its components, communication and environment.

Stored program: instructions and data share main memory in von Neumann architecture.

CPU: control unit coordinates, ALU calculates/compares and registers hold current values.

Instruction cycle: fetch → decode → execute → store; the shared instruction/data path can create a bottleneck.

Binary: base 2; each position is a power of 2. Addition carries 1 when a column totals 2 or 3.

Unsigned n-bit range: 0 to $2^n - 1$. **Signed two's-complement range:** -2^{n-1} to $2^{n-1} - 1$.

ASCII: standard 7-bit code for 128 characters. **Unicode:** worldwide code-point standard; UTF-8, UTF-16 and UTF-32 encode it.

Software: system software manages the platform; application software performs user tasks; utilities maintain performance and security.

Binary storage: 1 byte = 8 bits; the textbook uses 1 KB = 1024 bytes and 1024-based larger units.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	b	2	b	3	c	4	b	5	c
6	a	7	a	8	c	9	b	10	a

How to attempt this chapter in the paper

- For conversions, write the place values or repeated divisions clearly and read remainders in the correct direction.
- For binary arithmetic, align columns and show carries, borrows or shifted partial products. Finish with a decimal check.
- In architecture answers, draw and label the main blocks and explain both the stored-program idea and the instruction cycle.
- For a two's-complement negative number, state the fixed bit width, invert all bits and add 1—do not simply change the sign bit.

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